Section 3.1 Linear Transformations

A linear transformation is a very important concept in linear algebra and can be used to represent a wide variety of physical, economic, statistical and scientific problems. In this section we will study linear transforms that arise from the multiplication of a vector by a Matrix. Geometrically the action of the matrix on the vectors will be to shrink, stretch and reorient them.

A function $T: R^m \to R^n$ is a **linear transformation** if for all vectors $\vec{u}, \vec{v} \in R^m$ and all scalars r we have the following properties.

(a) $T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v})$
(b) $T(r\vec{u}) = rT(\vec{u})$

The properties given in (a) and (b) above are extremely useful in practice. Other things are often called linear as well and there is a whole theory on Linear Operators. The differentiation operator is one such operator. If you recall the sum of the derivatives is the derivative of the sum rule, that's (a). You might also remember the fact that when taking derivatives you can pull out a constant multiple, that's just property (b).

$$\frac{d}{dx}\big(f(x) + g(x)\big) = \frac{d}{dx}f(x) + \frac{d}{dx}g(x) \qquad \frac{d}{dx}cf(x) = c\frac{d}{dx}f(c)$$

Examples of transformations, are they linear?

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = x$$

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} \cos(x) \\ \sin(x) \end{bmatrix}$$

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 2x + 1 \\ xy \end{bmatrix}$$

$$T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} y + x \\ z - y \\ x \\ x + y + z \end{bmatrix}$$

$$T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} y^2 \\ x^2 \\ z \end{bmatrix}$$

Section 3.1 Linear Transformations

Linear Transformations as matrix multiplication

Let's revisit the linear transformation $T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} 2x+y \\ y-3z \\ -x \\ x+y+z \end{bmatrix}$, I clam we can write this in an equivlent form using a matrix times a vector. That is $T(\vec{u}) = A\vec{u}$.

Here in one line is the correspondence between the forms of our linear transformation.

$$T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} 2x+y \\ y-3z \\ -x \\ x+y+z \end{bmatrix} = \begin{bmatrix} 2x \\ 0 \\ -x \\ x \end{bmatrix} + \begin{bmatrix} y \\ y \\ 0 \\ y \end{bmatrix} + \begin{bmatrix} 0 \\ -3z \\ 0 \\ z \end{bmatrix} = x\begin{bmatrix} 2 \\ 0 \\ -1 \\ 1 \end{bmatrix} + y\begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix} + z\begin{bmatrix} 0 \\ -3 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & -3 \\ -1 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Theorem 3.2: Let A be an $n \times m$ matrix and $\vec{u} \in R^m$. Then $T(\vec{u}) = A\vec{u}$ is a linear transformation $T: R^n \to R^m$

Proof

Section 3.1 Linear Transformations

Let $T: R^m \to R^n$ be a a linear transformation $T(\vec{u}) = \vec{v}$.

We call the set of input vectors $\{\vec{u}\} \in R^m$ the **Domain** of T

We call the set of output vectors $\{\vec{v}\} \in R^n$ the **Codomain** of T

For a specific vector $\vec{u}$ in the domain of T we call $\vec{v} = T(\vec{u})$ the **Image** of $\vec{u}$ under T

The collection of the images of all of the vectors in the domain is called the **Range** of T

Example:
Define $T(\vec{u}) = \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$ Identify the Domain and Codomain and draw the Image of the vector $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$. What does the image of all of the vectors in the unit square $[0,1] \times [0,1]$ look like?

Section 3.1 Linear Transformations

Theorem 3.3: Let $A = [\vec{a}_1 \ \ \vec{a}_2 \ \ \cdots \ \ \vec{a}_m]$ be a $n \times m$ matrix, and let $T: R^m \to R^n$ with $T(\vec{x}) = A\vec{x}$ be a linear transformation. Then

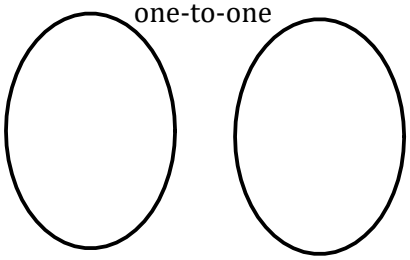
(a) The vector $\vec{w}$ is in the range of T if and only if $A\vec{x} = \vec{w}$ is a consistent linear system
(b) $range(T) = span\{\vec{a}_1, \vec{a}_2, \cdots, \vec{a}_m\}$

Proof of (b)

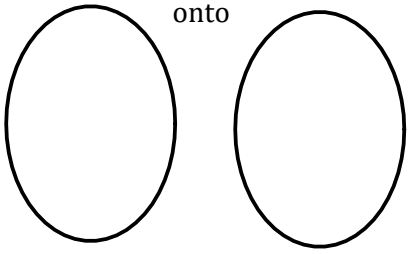
Let $T: R^m \to R^n$ be a linear transformation then

(a) T is one-to-one (injective) if for every vector $\vec{w} \in \mathrm{R}^n$ there exists at most one vector $\vec{u} \in \mathrm{R}^m$ such that $T(\vec{u}) = \vec{w}$.
(b) T is onto (surjective) if for every vector $\vec{w} \in R^n$ there exists at least one vector $\vec{u} \in \mathrm{R}^m$ such that $T(\vec{u}) = \vec{w}$

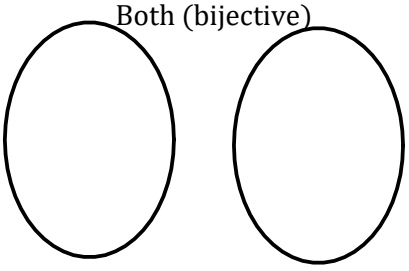
one-to-one



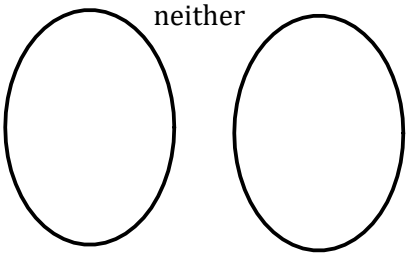
onto



Both (bijective)



neither



Section 3.1 Linear Transformations

Example: Assign the correct terminology to the linear transformations

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = x$$

$$T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} 2x \\ 2y \\ 2z \end{bmatrix}$$

$$T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} 2x - y \\ 2y - z \\ 2z - y \end{bmatrix}$$

$$T\left(\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix}\right) = \begin{bmatrix} 1 & 1 & 1 & -3 \\ -1 & 0 & -4 & 5 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 0 & 2 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix}$$

This one is a bit harder to tell off the bat, but we have a theorem that is helpful.

Theorem 3.5: Let T be a linear transformation. Then T is one-to-one if and only if the only solution to $T(\vec{x}) = 0$ is the trivial solution $\vec{x} = 0$.

Proof:

Suppose T is 1-1

Suppose the only solution to $T(\vec{x}) = 0$ is $\vec{x} = 0$.

Section 3.1 Linear Transformations

Let A be an $n \times m$ matrix and define $T: R^m \to R^n$by $T(\vec{x}) = A\vec{x}$. Then

(a) T is 1-1 if and only if the columns of A are linearly independent

(b) If $A \sim B$ where B is in echelon form then T is Is 1-1 if and only if B has a pivot position in every column.

(c) If $n < m$, then T is **not** 1-1

(d) T is onto if and only if the columns of A span the codomain R^n

(e) If $A \sim B$ and B is in echelon form, then T is onto if and only if B has a pivot position in every row

(f) If $n > m$, the T is **not** onto.

Theorem 3.9 The second unifying theorem

Let $S = \{\vec{a}_1, \vec{a}_2, \cdots, \vec{a}_n\}$ be a set of n vectors in R^n, Let $A = [\vec{a}_1 \cdots \vec{a}_n]$, and let $T: R^n \to R^n$ be given b $T(\vec{x}) = A\vec{x}$. Then the following are equivelent.

(a) $S\ spans\ R^n$

(b) $S\ is\ linearly\ independent$

(c) $A\vec{x} = \vec{b}\ has\ a\ solution\ for\ all\ b \in R^n$

(d) $T\ is\ onto$

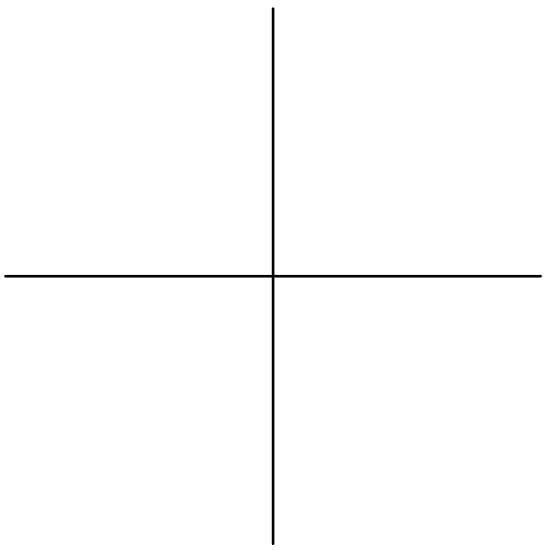
(e) $T\ is\ one - to - one$

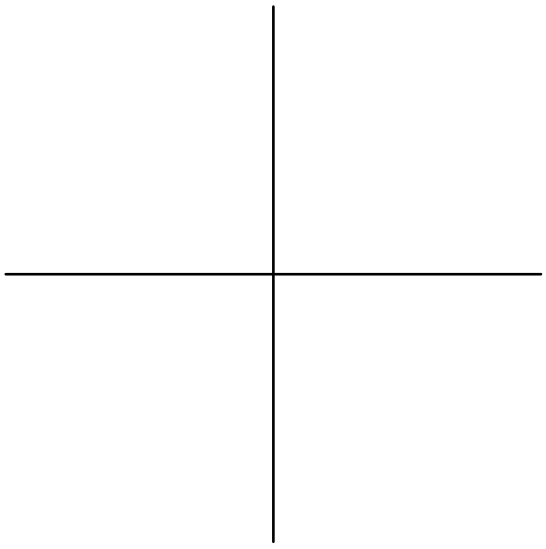
Section 3.1 Linear Transformations

Geometry of linear transformations. As we stated in the beginning linear transformations can shrink, stretch and reorient(rotate) vectors.

Scaling linear transformation $\begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix}$

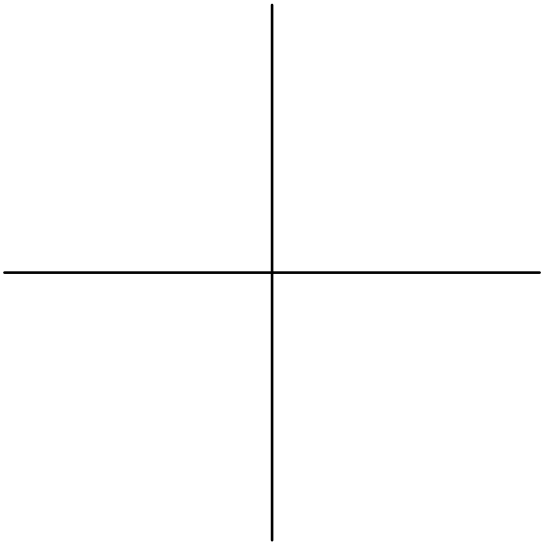
Ex Draw the before and after of the transformation $T\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix}$sketch what this does to the unit square

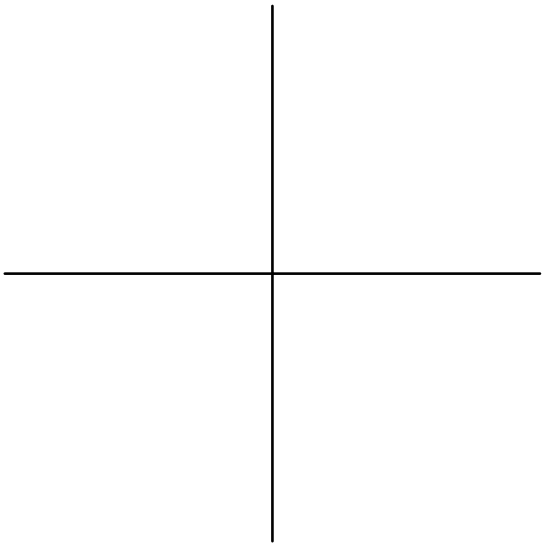




Shear linear transformation $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$

Ex Draw the before and after of the transformation $T\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix}$, sketch what this does to the unit square

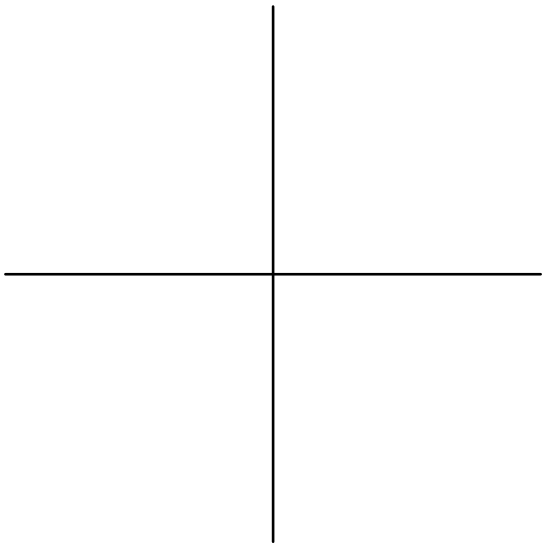


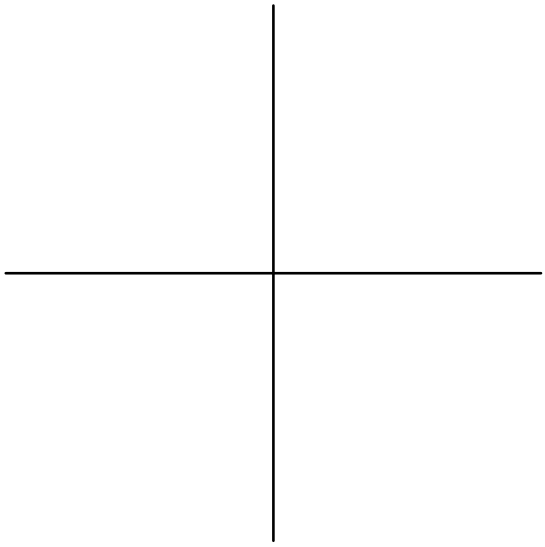


Section 3.1 Linear Transformations

Rotation $\begin{bmatrix} \cos(\alpha) & -\sin(\alpha) \\ \sin(\alpha) & \cos(\alpha) \end{bmatrix}$

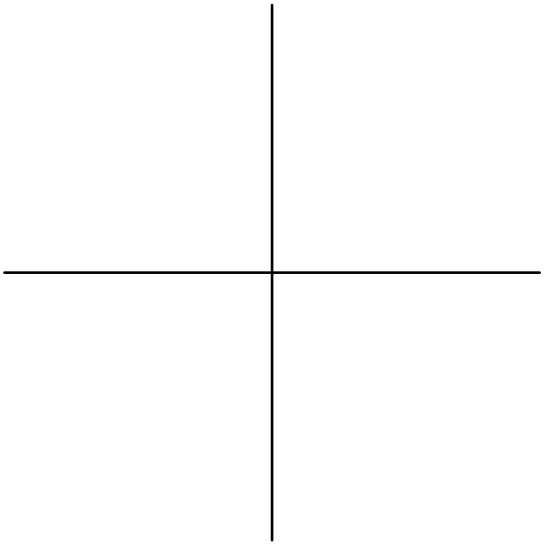
Ex Draw the before and after of the transformation $T\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ sketch what this does to the unit square

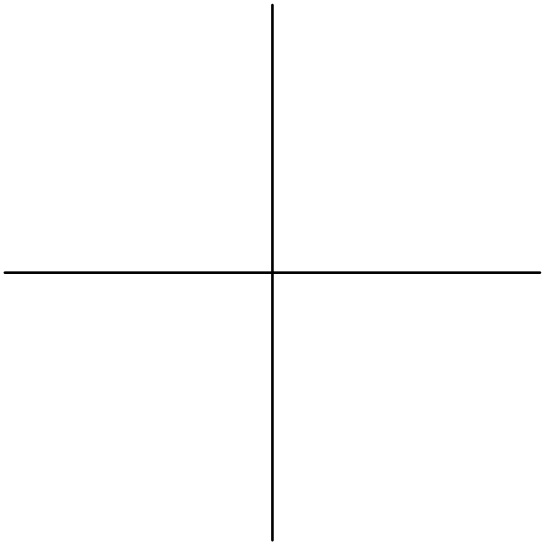




Reflection $\begin{bmatrix} \cos(2\alpha) & \sin(2\alpha) \\ \sin(2\alpha) & -\cos(2\alpha) \end{bmatrix}$

Ex Draw the before and after of the transformation $T\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix}$, sketch what this does to the unit square

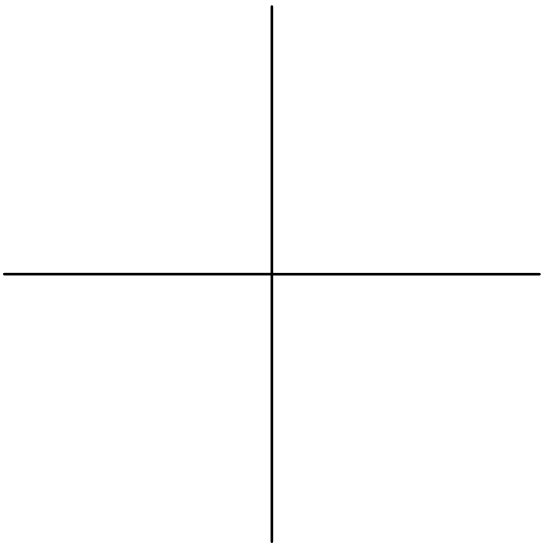


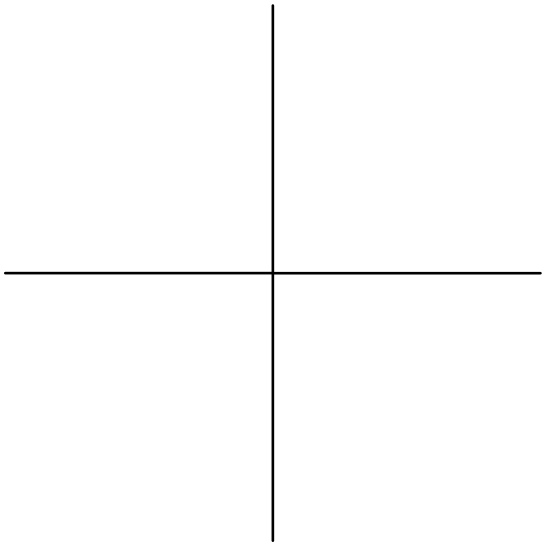


Section 3.1 Linear Transformations

Projection $\begin{bmatrix}1 & 0\\0 & 0\end{bmatrix}$ $\begin{bmatrix}0 & 0\\0 & 1\end{bmatrix}$

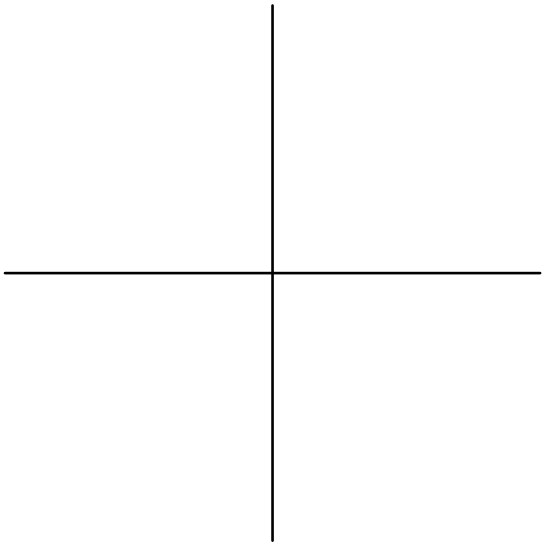
Ex Draw the before and after of the transformation $T\begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}1 & 0\\0 & 0\end{bmatrix}\begin{bmatrix}1\\1\end{bmatrix}$, sketch what this does to the unit square

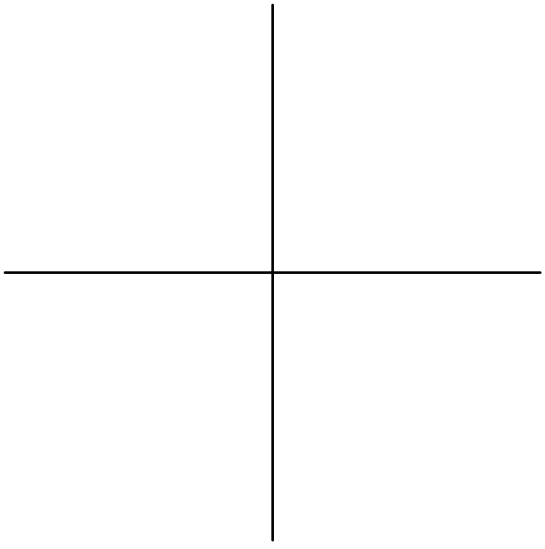




Rotation dilatation $\begin{bmatrix}a & -b\\b & a\end{bmatrix}$

Ex Draw the before and after of the transformation $T\begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}2 & -1\\1 & 2\end{bmatrix}\begin{bmatrix}1\\1\end{bmatrix}$, sketch what this does to the unit square





Section 3.1 Linear Transformations

Combining linear transformations.

Ex: Find a linear transformation that rotates a vector by 90^0 , scales it by 2, and shears it.

Apply the transformation the the vector $\begin{bmatrix}2\\1\end{bmatrix}$ and sketch it and it's image.

